Forensic Research Project – Project Planning Document

Experimental Factor	Information
Group Members	Completed as part of a UTS group project. Group member names and personal details have been removed from this public version.
Practical Session Time	Friday, 1PM
Project Title	Developing a Methodological Framework for Investigating Online Extremism Communities on the Clear Web
Project Aims	- To develop a framework to find specific information relating to a case and store and label it ethically.
Project Objectives	• Identify clear web platforms hosting publicly accessible community forums • Evaluate tools and methods for the acquisition of online text data under platform and ethical constraints
Hypotheses	- Web based scrapers will be less efficient and collecting relevant data then programmed software created on python -
Relevant literature	https://dl.acm.org/doi/epdf/10.1145/2207676.2208610 https://www.sciencedirect.com/science/article/pii/S1359178921000628 chrome-extension://kdpelmjpfafjppnhbloffcjpeomlnpah/https://aclanthology.org/W12-2103.pdf
Key points from literature	
Internet, social media and online hate speech. Systematic review https://www.sciencedirect.com/science/article/pii/S1359178921000628	Aim: explore research papers related to how Internet and social media may, or may not, constitute an opportunity to online hate speech The following inclusion criteria were employed to identify eligible studies: 1. any paper that addresses the relationship between Internet use or social networks and online hate speech or cyberhate, except annotations, conferences, narrative or systematic reviews, letters and comments;

	2. published between January of 2015 and September 2019 given the fast changes in the last five years in the development of Web 2.0 applications that are remarkable even set against the pace of change since the advent of the Internet; 3. focused on any specific type of online hate speech 4. written in English or Spanish, not discriminating by geographical area; 5. there were no restrictions for cross-sectional or longitudinal designs. Types of online hate 1. Online religious hate speech 2. Online racism 3. Political online hate 4. Gendered online hate Computational methods or software such as Apache Spark, Python packages, including Tweepy, UCINET and NodeXL among others; through the search for keywords related to hate speech intend to analyze and create algorithms, by combining with quantitative analysis methods to obtain more reliable scales and reduce misinterpretation of the results obtained.
Detecting Hate Speech on the World Wide Web https://aclanthology.org/W12-2103.pdf	Aim: approach to detecting hate speech in online text, where hate speech is defined as abusive speech targeting specific group characteristics, such as ethnic origin, religion, gender, or sexual orientation. Approach: used the template-based strategy to generate features from the corpus. Included features such as surrounding words around target word (2 word window), part-of speech tagging, word clusters, words within 10 word window, and associations between words and hate labels Data: news group posts and comments from Yahoo! that were flagged as offensive by readers. Observed evasion tactics in comments such as misspelling words “nagger”, using homphones “joo”, and adding symbols “j@e@w”. Some hate speech did not contain slurs and were formatted as a scientifically worded essays.

	Errors: false negatives (failed detection of hate speech) and false positives (wrongly flagged) did occur in the research
Collecting Data on Textiles from the internet using web scraping tools - [Collecting data on textiles from the internet using web crawling and web scraping tools - ProQuest]()	Aim: to use web scraping tools on clothing websites to get accurate, real-time data on what are the most common fabrics and colors being sold to identify clothing samples. The study used a bot crawler to browse and a bot scraper to collect data from the internet. These tools allow data to be obtained in a systematic and automated way. The scraper and browser bot were able to collect information on 24,000 different clothes including their composition material, colour and size. The study used Scrapy 1.5.2 and Elasticsearch 6.7.0. Scrapy was used to write the data collection application and Elasticsearch was used for data storage and acquisition monitoring. Kibana was also used to provide real-time monitoring of collected data and give a visual interface with live statistics. Steps: 1. Analyze a target website for structure and isolate the desired information. 2. Use a data collection application (Scrapy) 3. Use data storage/cleaning application (KNime? Elasticsearch)
Dinkins, J., Pascual, M. G., Aguilera, M., Bothwell, S., Schmiege, S., Afrin, A., Prose, N., & Kohn, L. L. (2024). Representation of Racially Minoritized Patients on Dermatology Private Practice Websites. *Journal of Racial and Ethnic Health Disparities*. https://doi.org/10.1007/s40615-024-02061-6	- Apify being used to get dental office locations off of Google Maps
Jahanbin, K., & Chahooki, M. A. Z. (2024). Database of twitter influencers in cryptocurrency (2021–2023) with	- X's API was used on Apify to create a database of 52 users and what their opinions on Crypto currencies were.

sentiments. *BMC Research Notes*, *17*(1), Article 303. https://doi.org/10.1186/s13104-023-06548-z	
Exploring Dark Web Crawlers: A Systematic Literature Review of Dark Web Crawlers and Their Implementation – DOAJ	
Consumables/Equipment Needed	Python PC Mouse Keyboard Monitors Knime
Control	- Websites/information consumed by different frameworks? - The terms that are searched with the frameworks - Hardware and environments used - Data storage formats (JSON) - Case topic (Schapelle Corby)
Constants	- Ethical framework followed (only using publicly available data) - No interaction with users/communities - Anonymization of data - Text preprocessing methodology (tokenization and clearing)
Replicates	- Each scraping method will run the same amount of times - Each run will use the same search queries
Independent variable/s	- Scraping methods used (i.e. Beautiful Soup, Scapy, Selenium)
Dependent variable	- Volume of data that is collected (number of posts/comments/images) - Processing success (% of pages/documents successfully collected without error) - Time efficiency
Number of Specimens	N/A
Brief Experimental Method	1. Identify relevant and accessible clear-web platforms and their regulations that discuss our case study 2. Define the standard search terms to be used for the search queries 3. Identify and apply the best scraping methods for the task 4. Collect a fixed number of posts/comments from websites scraped

	5. Store data in a consistent format (JSON) 6. Preprocess text (cleaning, normalization, tokenization, etc.) 7. Evaluate results by dependent variables 8. Compare the scraping methods used and document comparison metrics
Other Considerations	- **Ethical Considerations** – only collecting publicly available data. Remove personal identification of comments collected. Respect platform T&S/robots.txt. Data should be stored securely and anonymized. No interaction with individuals on platforms.
Analysis Method	
Case Scenario	
Schapelle Corby How the Schapelle Corby case unfolded https://www.9news.com.au/national/how-the-schapelle-corby-case-unfolded/329745ae-c343-420c-bcaf-56260148dd84	October 8, 2004 - Arrested at Bali airport with 4.1kg of cannabis in bodyboard bag. January 27, 2005 - Corby's trial begins. She accuses police of ignoring crucial evidence that could prove her innocence. April 21, 2005 - Prosecutors call for Corby to be jailed for life. May 27, 2005 - Found guilty of drug trafficking and sentenced to 20 years' jail. August 9, 2009 - Australian psychiatrist Jonathan Phillips visits Corby in prison and states she is mentally ill, recommending she be transferred to an Australian prison for treatment. May 22, 2012 - Indonesian President Susilo Bambang Yudhoyono orders a five-year reduction in sentence. December 25, 2013 - Sentence cut by two months as part of a Christmas remission program. February 10, 2014 - Corby is released on parole May 27, 2017 - Corby set to report to correction officials for the last time in Bali before being deported to Australia.

Experimental Timeline

Timeline	Activity/ Experiments
Week 1	No Practical Activity
Week 2	• Risk Assessment • Project Planning
Week 3	• Research on hate speech terms, scraping methods, ethics, terms relating to Chapelle Corby • Begin constructing slides
Week 4	• **Project Plan Presentation (Workshop)** • Consolidate research papers and prior data from hypothesis, aim & objectives, etc
Week 5 & 6	• Testing and applying free and online preprogrammed scraping tools whilst stating both positives and negatives of each • Planning initial approach towards the use of python/Knime code to create our own scraping tool
Week 7 & 8	• Applying and using programmed and api scrapers and collecting relevant data for the final report • Begin constructing slides for prject update with some refined data
STUVAC	No Classes Scheduled
Week 9	• **Project Update Presentation (Workshop)** • Begin to properly use chosen scrapers to collate and consolidate data
Week 10 & 11	• Collate, Consolidate and finalise data
Week 12	• Analyse results and compare conclusions as well as explore alternatives through reflection of project • **Final Presentation (Workshop)**

Wants and hopes on what we want to do. Stuff we find in articles and how it could benifit the project.

Links to Literature Resources:

Need a paper on common examples of hate speech (words)

Need to decide on the platform we are getting the information from.

Artificial Intelligence in Forensic Sciences: A Systematic Review of Past and Current Applications and Future Perspectives - PMC

Data collection methods:

Website Scraping with Python: Using BeautifulSoup and Scrapy

TUTORIAL: AI research without coding: The art of fighting without fighting: Data science for qualitative researchers
https://www.sciencedirect.com/science/article/pii/S0148296320303854

Ethics:

Applications and Challenges of Artificial Intelligence for Digital Forensics – Cyber

A framework for ethical AI

Hate Speech Intensity Scale:

https://www.ssoar.info/ssoar/bitstream/handle/document/86421/ssoar-2023-bahador-Monitoring_hate_speech_and_the.pdf?sequence=1&isAllowed=y